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BOND PAD STRUCTURE WITH REDUCED STEP HEIGHT AND INCREASED ELECTRICAL ISOLATION

Abstract

Various embodiments of the present disclosure are directed towards an integrated chip. The integrated chip includes a substrate having an upper surface vertically below a top surface. A conductive structure is in the substrate. The conductive structure includes a first segment overlying the upper surface of the substrate and one or more vertical segments under the first segment. A first isolation structure is in the substrate and on opposing sides of the conductive structure. A second isolation structure is in the substrate and around the one or vertical segments. A top surface of the first isolation structure and a top surface of the second isolation structure are vertically below the top surface of the substrate. A bottom surface of the first isolation structure and a bottom surface of the second isolation structure are vertically above a bottom surface of the substrate.

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Background/Summary

REFERENCE TO RELATED APPLICATIONS [0001] This Application is a Continuation of U.S. application Ser. No. 17/555,896, filed on Dec. 20, 2021, which is a Divisional of U.S. application Ser. No. 16/558,556, filed on Sep. 3, 2019 (now U.S. Pat. No. 11,217,547, issued on Jan. 4, 2022). The contents of the above-referenced Patent Applications are hereby incorporated by reference in their entirety.

BACKGROUND
[0002] Integrated circuits (ICs) with image sensors are used in a wide range of modern day electronic devices, such as cameras and cell phones, for example. Complementary metal-oxide semiconductor (CMOS) devices have become popular IC image sensors. Compared to charge-coupled devices (CCD), CMOS image sensors are increasingly favored due to low power consumption, small size, fast data processing, a direct output of data, and low manufacturing cost. Some types of CMOS image sensors include front-side illuminated (FSI) image sensors and back-side illuminated (BSI) image sensors.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.
[0004] FIG. **1** illustrates a cross-sectional view of some embodiments of an integrated chip (IC) including a bond pad extending through a semiconductor substrate.
[0005] FIGS. **2**A-**2**B illustrate top views of some alternative embodiments of the IC of FIG. **1** taken along the line A-A′.
[0006] FIG. **3** illustrates a cross-sectional view of some embodiments of an image sensor including a bond pad laterally offset from a plurality of photodetectors.
[0007] FIG. **4** illustrates a top view of some embodiments of a semiconductor structure including multiple bond pads laterally surrounding device regions.
[0008] FIG. **5** illustrates a cross-sectional view of an IC according to some alternative embodiments of the IC of FIG. **1**.
[0009] FIGS. **6-16** illustrate a series of cross-sectional views of some embodiments of a method for forming a bond pad with a reduced step height and a bond pad isolation structure surrounding the bond pad.
[0010] FIG. **17** illustrates a block diagram of some embodiments of the method of FIGS. **6-16**.
DETAILED DESCRIPTION
[0011] The present disclosure provides many different embodiments, or examples, for implementing different features of this disclosure. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and

second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0012] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0013] Some complementary metal-oxide-semiconductor (CMOS) image sensors include an array of pixels sensors arranged within a pixel array region of a substrate of an integrated circuit (IC). The pixel sensors each include a plurality of photodetectors arranged within the substrate in proximity to a back-side of the substrate, so that the photodetectors are able to receive light along the back-side of the substrate. An array of color filters is arranged over the pixel sensor array and is buried within a dielectric layer that overlies the back-side of the substrate. Burying the color filter array within the dielectric layer advantageously improves the quantum efficiency (QE) of the image sensors. Bond pads are disposed within a bond pad region that is a peripheral region of the substrate laterally offset from the pixel array region. The bond pads are formed before performing the buried color filter array (BCFA) process. For example, a bond pad is formed within a bond pad opening and is vertically offset from the back-side of the substrate by a step height. Forming the bond pad opening may include removing an entire thickness of the substrate above a shallow trench isolation structure (STI), such that the bond pad does not extend through the substrate. The buried color filter array (BCFA) process is carried out after forming the bond pad. For example, the dielectric layer is formed over the back-side of the substrate and fills a remaining portion of the bond pad opening. Subsequently, the array of color filters may be formed in and/or over the dielectric layer in a region laterally offset from the bond pad.

[0014] In some embodiments, challenges with the aforementioned structure arise as the thickness of the substrate increases. As the thickness of the substrate increases (e.g., to a range of about 3.5-6 micrometers) an intrinsic absorption coefficient of the substrate material (e.g., silicon) increases, thereby increasing the quantum efficiency (QE) of the image sensor. However, as the substrate thickness increases, so does the step height (e.g., from about 2.75 to about 5.75 micrometers) between the bond pad and the back-side of the substrate. Because of the increased step height, the dielectric layer within the bond pad opening is unable to uniformly fill gaps around and/or over the bond pad (e.g., due to a limitation in tools used to form the dielectric layer). This results in cracking and/or delamination of layers adjacent to the bond pad. Additional processing steps (e.g., additional dielectric depositions and/or patterning processes) may be carried out to fill the gaps, however this increases time and costs associated with fabricating the IC. Further, by disposing the bond pad over the shallow trench isolation (STI) structure, such that the bond pad does not extend through the substrate, noise in the image sensor may be reduced, but this will contribute to the increased step height.

[0015] Accordingly, in some embodiments, the present disclosure relates to a bond pad with a decreased step height and an associated method for forming the bond pad that simplifies the process for filling a bond pad opening. In some embodiments, a passivation layer is formed over a back-side of a substrate. The passivation layer and the substrate are patterned to form the bond pad opening. The patterning process defines an upper surface of the substrate vertically below the back-side. The bond pad opening includes bond pad protrusion openings that extend through an entire thickness of the substrate and expose an upper surface of a metal line within an underlying interconnect structure. A first bond pad isolation structure is formed in the substrate around the bond pad protrusion openings, and a second bond pad isolation structure is formed along sidewalls of the substrate that define the bond pad protrusion openings. Subsequently, a bond pad is formed in the bond pad opening, such that the bond pad has an upper conductive body overlying the upper surface of the substrate and conductive protrusions extending from the conductive body to the metal line. Subsequently, the dielectric layer is formed over the passivation layer and fills a remaining portion of the bond pad opening. By forming the bond pad over the upper surface of the substrate, the step height is reduced (e.g., to about 0 to 1.80 micrometers) and a thickness of the dielectric layer in the bond pad opening is reduced. This, in part, may mitigate and/or eliminate cracking and/or delamination of layers adjacent to the bond pad. Further, the first and second bond pad isolation structures separate the bond pad from the substrate, thereby decreasing and/or eliminating “leakage” between the bond pad and the photodetectors. This may decrease noise present in the image sensor.

[0016] FIG. **1** illustrates a cross-sectional view of some embodiments of an integrated chip **100** having a bond pad **116** disposed within a bond pad region **101***b*.

[0017] The integrated chip **100** includes an interconnect structure **104** disposed along a front-side **110***f* of a semiconductor substrate **110** (e.g., a silicon substrate). The interconnect structure **104** overlies a carrier substrate **102** (e.g., a silicon substrate), where the interconnect structure **104** is disposed between the semiconductor substrate **110** and the carrier substrate **102**. The interconnect structure **104** includes a plurality of interconnect layers arranged within an interconnect dielectric structure **103**. The plurality of interconnect layers alternate between conductive wires **108** and conductive vias **106**. The conductive wires **108** are configured to provide a lateral connection (i.e., a connection parallel to an upper surface of the carrier substrate **102**), whereas the conductive vias **106** are configured to provide for a vertical connection between adjacent conductive wires **108**. The conductive wires **108** include an upper conductive wire layer **108***a*. A passivation structure **118** overlies a back-side **110***b* of the semiconductor substrate **110**. A first dielectric layer **120** overlies the passivation structure **118**. A shallow trench isolation (STI) structure **112** is disposed within the semiconductor substrate **110** and extends along an upper surface of the interconnect dielectric structure **103**.

[0018] The bond pad region **101***b* extends through the semiconductor substrate **110**, at a location laterally offset from a device region **101***a*, to the upper conductive wire layer **108***a*. In some embodiments, the device region **101***a* includes one or more semiconductor devices **126** (e.g., transistor(s), resistor(s), varactor(s), etc.) and/or photodetectors (not shown) disposed within and/or on the semiconductor substrate **110**. The bond pad **116**, a first bond pad isolation structure **114**, and second bond pad isolation structures **115** are disposed within the bond pad region **101***b*. The bond pad **116** includes an upper conductive body **116***a* and conductive protrusions **116***b* underlying the upper conductive body **116***a*.

[0019] The upper conductive body **116***a* comprises a same material (e.g., aluminum copper) as the conductive protrusions **116***b*. The upper conductive body **116***a* overlies the upper surface **110***us* of the semiconductor substrate and is separated from the upper surface **110***us* by the passivation structure **118**. Further, the conductive protrusions **116***b* continuously extend through the semiconductor substrate **110**, the STI structure **112**, and the interconnect dielectric structure **103**. Furthermore, the upper conductive body **116***a* has sidewalls that define bond pad openings **1160** overlying the conductive protrusions **116***b*. An electrical connector pad **122** is disposed laterally between the bond pad openings **1160** and provides a wire bonding location for a conductive wire **124**. In some embodiments, the conductive wire **124** is bonded to another integrated chip (not shown), where the interconnect structure **104** is electrically coupled to the another integrated chip by way of the bond pad **116**. A conductive ring structure **113** is disposed over the upper surface **110***us* of the semiconductor substrate **110** and laterally surrounds the upper conductive body **116***a*.

[0020] The first bond pad isolation structure **114** is laterally offset from and surrounds outer sidewalls of the bond pad **116**. In some embodiments, the first bond pad isolation structure **114** comprises a material (e.g., silicon dioxide) different from the semiconductor substrate **110**. Thus, the first bond pad isolation structure **114** is configured to increases electrical isolation between the bond pad **116** and other devices (e.g., the semiconductor devices **126**) disposed on and/or within the semiconductor substrate **110**. Further, the second bond pad isolation structures **115** respectively surround and/or directly contact an adjacent conductive protrusion **116***b*. The second bond pad isolation structures **115** each extend from the upper conductive body **116***a* to the STI structure **112**. This further increases electrical isolation between the bond pad **116** and other devices disposed on and/or within the semiconductor substrate **110**. By virtue of the first bond pad isolation structure **114** and/or the second bond pad isolation structures **115** surrounding the bond pad **116**, "leakage" (i.e., a flow of current) between the bond pad **116** and the other devices and/or doped regions disposed within and/or on the semiconductor substrate **110** may be mitigated and/or eliminated. This, in part, may increase a reliability and/or endurance of the integrated chip **100**.

[0021] A top surface of the bond pad **116** is vertically offset from the back-side **110***b* of the semiconductor substrate **110** by a step height h**1**. In some embodiments, the step height h**1** is within a range of 0 to 1.80 micrometers. In some embodiments, if the step height h**1** is greater than 0 micrometers, then a thickness of the first dielectric layer **120** overlying the bond pad **116** may be decreased, such that the first dielectric layer **120** may uniformly fill gaps around and/or over the bond pad **116** during fabrication of the integrated chip **100**. This increases reliability and/or endurance of the integrated chip **100**. In further embodiments, if the step height h**1** is less than 0 micrometers, then the bond pad **116** may adversely interact with electromagnetic radiation (e.g., reflect the electromagnetic radiation) disposed upon the back-side **110***b* of the semiconductor substrate **110**. For example, this may decrease sensitivity of photodetectors disposed within the device region **101***a* of the semiconductor substrate **110** and/or increase noise present in the integrated chip **100**. In yet further embodiments, if the step height h**1** is greater than 1.80 micrometers, then the first dielectric layer **120** may not uniformly fill gaps around and/or over the bond pad **116**, such that the unfilled gaps may cause cracking and/or delamination of layers adjacent to the bond pad **116**.

[0022] In some embodiments, a lower surface of the upper conductive body **116***a* has a second height h**2**, as measured from the back-side **110***b* of the semiconductor substrate **110**. The second height h**2** is greater than the step height h**1**. The upper surface **110***us* of the semiconductor substrate **110** has a third height h**3**, as measured from the back-side **110***b* of the semiconductor substrate **110**. The third height h**3** is greater than the step height h**1**. Further, the lower surface of the conductive protrusions **116***b* has a fourth height h**4**, as measured from the back-side **110***b* of the semiconductor substrate **110**. The fourth height is greater than the step height h**1** and is greater than a thickness Ts of the semiconductor substrate **110**. In further embodiments, the top surface of the bond pad **116** may be aligned with the back-side **110***b* of the semiconductor substrate **110** (e.g., the step height h**1** may be zero). In yet further embodiments, the top surface of the bond pad **116** may not be disposed above the back-side **110***b* of the semiconductor substrate **110** (e.g., the step height h**1** may not be negative). In some embodiments, a thickness Tcb of the upper conductive body **116***a* is about 1.2 micrometers.

[0023] FIG. **2**A illustrates a top view of some alternative embodiments of the integrated chip **100** taken along line A-A′ of FIG. **1**. For ease of illustration, the first dielectric layer **120** of FIG. **1** has been omitted from the top view of FIG. **2**A.

[0024] The first bond pad isolation structure **114** has a ring like shape, where sidewalls of the first bond pad isolation structure **114** completely surround outer sidewalls of the bond pad **116**. When viewed from above, the first bond pad isolation structure **114** has a rectangular/square shape with rounded edges, however the first bond pad isolation structure **114** may have other shapes, such as a circular/elliptical shape. When viewed from above, the bond pad **116** has a rectangular/square shaped with rounded etches, however the bond pad **116** may have other shapes, such as a circular/elliptical shape. In some embodiments, a solder bump (not shown) may be disposed laterally between the bond pad openings **1160**.

[0025] When viewed from above, the bond pad openings **1160** may, for example, have a square/rectangular shape. The second bond pad isolation structures **115** completely surrounds an outer perimeter of the conductive protrusions (**116***b* of FIG. **1**) of the bond pad **116**. In such embodiments, the second bond pad isolation structures **115** may each have a shape that corresponds to an adjacent bond pad opening **1160**. Thus, the second bond pad isolation structures **115** separate the bond pad **116** from the semiconductor substrate **110**, thereby electrically isolating the bond pad **116** from other semiconductor devices and/or doped regions disposed within and/or on the semiconductor substrate **110**. However, in some embodiments, a thickness of the second bond pad isolation structure **115** may not be substantially thick enough such that “leakage” may occur between the bond pad **116** and the semiconductor substrate **110**. In such embodiments, the first bond pad isolation structure **114** further increases the electrical isolation of the bond pad **116** from the semiconductor substrate **110**. This, in part, mitigates and/or eliminates “leakage” between the bond pad **116** and the other semiconductor devices and/or doped regions disposed within and/or on the semiconductor substrate **110**. A shown in the cross-sectional view of FIG. **1**, the STI structure **112** directly underlies the first bond pad isolation structure **114**. In some embodiments, outer sidewalls of the bond pad **116** and/or outer sidewalls of the first bond pad isolation structure **114** are each laterally spaced between outer sidewalls of the STI structure **112**. In further embodiments, the first bond pad isolation structure **114** is disposed around a center of the bond pad **116**.

[0026] In some embodiments, the bond pad **116** may have a length L**1** and a width W**1**. The length L**1** may, for example, be within a range of about 75 to 85 micrometers. The width W**1** may, for example, be within a range of about 85 to 95 micrometers. In some embodiments, the width W**1** is greater than the length L**1**. In some embodiments, the bond pad openings **1160** may each have a length L**2** and a width W**2**. The length L**2** may, for example, be within a range of about 15 to 25 micrometers. The width W**2** may, for example, be within a range of about 2 to 8 micrometers. In some embodiments, the length L**2** is greater than the width W**2**.

[0027] FIG. **2**B illustrates a top view of some alternative embodiments of the integrated chip **100** taken along line A-A′ of FIG. **1**. For ease of illustration, the first dielectric layer **120** of FIG. **1** has been omitted from the top view of FIG. **2**B.

[0028] As illustrated in FIG. **2**B, an inner sidewall of the first bond pad isolation structure **114** is in direct contact with a sidewall of at least one of the conductive protrusions (**116***b* of FIG. **1**) of the bond pad **116**. Further, outer sidewalls of the first bond pad isolation structure **114** are laterally spaced between outer sidewalls of the bond pad **116**.

[0029] FIG. **3** illustrates a cross-sectional view of some embodiments of an image sensor **300** that includes a bond pad **116** laterally offset from a plurality of photodetectors **328**.

[0030] A semiconductor substrate **110** overlies an application specific integrated circuit (ASIC) substrate **301**. In some embodiments, the semiconductor substrate **110** and/or the ASIC substrate **301** may respectively, for example, be a bulk substrate (e.g., a bulk silicon substrate), a silicon-on-insulator (SOI) substrate, a silicon germanium (SiGe) substrate, or some other suitable substrate. A plurality of transistors **302** are disposed over the ASIC substrate **301**. The transistors **302** each comprise a gate electrode **310**, a gate dielectric **306**, a sidewall spacer structure **308**, and source/drain regions **304**. An interconnect structure **104** and an ASIC interconnect structure **312** are disposed between the semiconductor substrate **110** and the ASIC substrate **301**. The interconnect structure **104** and the ASIC interconnect structure **312** each comprise an interconnect dielectric structure **103**, a plurality of conductive wires **108**, and a plurality of conductive vias **106**. The interconnect structure **104** and the ASIC interconnect structure **312** are configured to electrically couple the transistors **302** to the photodetectors **328** and/or transfer transistors **320** disposed in the semiconductor substrate **110**. In some embodiments, the transfer transistors **320** each comprise a dielectric transfer layer **324** overlying a transfer electrode **322**. The semiconductor substrate **110** is bonded to the ASIC substrate **301** by way of the interconnect structure **104** and the ASIC interconnect structure **312**. In some embodiments, the conductive wires **108** and/or the conductive vias **106** may respectively, for example, be or comprise aluminum, copper, aluminum copper, tungsten, or the like. In some embodiments, the interconnect dielectric structure **103** may, for example, comprise one or more dielectric layers (e.g., silicon dioxide). In further embodiments, the ASIC substrate **301** may be configured as the carrier substrate **102** of FIG. **1**. In such embodiments, the ASIC interconnect structure **312** may be omitted.

[0031] The plurality of photodetectors **328** are disposed within the semiconductor substrate **110**. In some embodiments, the photodetectors **328**, the transistors **302**, and the transfer transistors **320** are laterally spaced within the device region **101***a*. The semiconductor substrate **110** may have a first doping type (e.g., p-type) and the photodetectors **328** may each have a second doping type (e.g., n-type) opposite the first doping type. In some embodiments, the photodetectors **328** respectively extend from a back-side **110***b* of the semiconductor substrate **110** to a point below the back-side **110***b*. In further embodiments, the point is at a front-side **110***f* of the semiconductor substrate **110**, which is opposite the back-side **110***bs* of the semiconductor substrate **110**. In some embodiments, isolation structures **325**, **326** may be disposed within the semiconductor substrate **110** laterally disposed between adjacent photodetectors **328**. For example, a shallow trench isolation (STI) structure **326** may be disposed into the front-side **110***f* of the semiconductor substrate **110** to a point above the transfer transistors **320**. The isolation structures **326** may be configured as STI structures and may, for example, be or comprise silicon dioxide, silicon nitride, or the like. Further, an elongated isolation structure **325** may extend from the isolation structures **326** to the back-side **110***b* of the semiconductor substrate **110**. The elongated isolation structures **325** may be configured as deep trench isolation (DTI) structures and/or may be doped regions of the semiconductor substrate **110** configured to electrically isolate the photodetectors **328** from one another.

[0032] The photodetectors **328** are each configured to convert electromagnetic radiation (e.g., photons) into electrical signals (e.g., to generate electron-hole pairs from the electromagnetic radiation). In some embodiments, the photodetectors **328** may, for example, each be configured to generate electrical signals from near infrared radiation (NIR) electromagnetic radiation (e.g., wavelength of about 0.7-5 micrometers). In such embodiments, a material and/or a thickness Ts of the semiconductor substrate **110** is configured to ensure the photodetectors **328** have high quantum efficiency (QE) in NIR applications. For example, the thickness Ts of the semiconductor substrate **110** may be within a range of about 3.5 to 6 micrometers. In some embodiments, if the thickness Ts is greater than about 3.5 micrometers, then the photodetectors **328** will each have enhanced NIR light QE, and this may have an increased ability for phase detection and/or depth detection. In further embodiments, if the thickness Ts is less than about 6 micrometers, then the photodetectors **328** will each have enhanced NIR light QE while mitigating complexity, costs, and time associated with fabricating the image sensor **300**. In yet further embodiments, the photodetectors **328** may, for example, each be configured to generate electrical signals from visible light (e.g., wavelength of about 0.38 to 0.75 micrometers).

[0033] The passivation structure **118** overlies the back-side **110***b* of the semiconductor substrate **110** and comprises a first passivation layer **118***a*, a second passivation layer **118***b*, and a third passivation layer **118***c*. The first passivation layer **118***a* may, for example, be or comprise an oxide, such as silicon dioxide, a low-k dielectric material, or the like. The first passivation layer **118***a* may directly contact the bond pad **116**. The second passivation layer **118***b* may, for example, be or comprise an oxide, such as silicon dioxide, a low-k dielectric material, or the like. The third passivation layer **118***c* may, for example, be or comprise a high-k dielectric material, or some other suitable dielectric material. The passivation structure **118** may be configured to protect the back-side **110***b* of the semiconductor substrate **110**.

[0034] A grid structure **329** overlies the passivation structure **118**. The grid structure **329** includes a first grid layer **330** and a second grid layer **332** overlying the first grid layer **330**. The grid structure **329** is laterally around and between the photodetectors **328** to define a plurality of color filter openings. A dielectric protection layer **334** may be disposed along an upper surface and sidewalls of the grid structure **329**. In some embodiments, the dielectric protection layer **334** may be configured as a sidewall protection structure that prevents damage to the grid structure **329**. The dielectric protection layer **334** may, for example, be or comprise an oxide, such as silicon dioxide, or another suitable dielectric material. A plurality of light filters **336** are arranged within the plurality of color filter openings and overlies the plurality of photodetectors **328**. The grid structure **329** may comprise a dielectric material with a refractive index less than a refractive index of the light filters **336**. Due to the lower refractive index, the grid structure **329** serves as a radiation guide to direct incident electromagnetic radiation (e.g., NIR light) to a corresponding photodetector **328**. Further, the light filters **336** are each configured to block a first range of frequencies of the incident electromagnetic radiation while passing a second range of frequencies (different than the first range of frequencies) of the incident electromagnetic radiation to an underlying photodetector **328**. In some embodiments, the second range of frequencies may be NIR light. In yet further embodiments, the plurality of color filters may be configured as a buried color filter array (BCFA), such that the second range of frequencies may be visible light. Further, a plurality of micro-lenses **338** may be disposed over the plurality of light filters **336**.

[0035] The device region **101***a* is laterally offset from a bond pad region **101***b* of the image sensor **300**. A bond pad **116** and a first bond pad isolation structure **114** are laterally arranged within the bond pad region **101***b* of the image sensor **300**. Thus, in some embodiments, the bond pad **116** and the first bond pad isolation structure **114** are laterally offset from the photodetectors **328** and/or the transfer transistors **320** by a non-zero distance.

[0036] The bond pad **116** is configured to electrically coupled the transfer transistors **320**, the transistors **302**, and/or the photodetectors **328** to another integrated chip (not shown) by way of the interconnect structure **104** and/or the ASIC interconnect structure **312**. In some embodiments, the bond pad **116** is separated from an upper surface of the semiconductor substrate **110** by the first passivation layer **118***a* and has protrusions extending through the semiconductor substrate **110** to the upper conductive wire layer **108***a*. In such embodiments, the protrusions are separated from sidewalls of the semiconductor substrate **110** by a second bond pad isolation structure **115**. The protrusions extend through a shallow trench isolation (STI) structure **112** and the interconnect dielectric structure **103**. Further, the bond pad **116** has sidewalls that define bond pad openings **1160** overlying the protrusions of the bond pad **116**. In some embodiments, a height of the bond pad **116** is greater than the thickness Ts of the semiconductor substrate **110**. Further, a first dielectric layer **120** is disposed over an upper surface of the bond pad **116** and may have a substantially flat upper surface that is aligned with an upper surface of the first passivation **118***a*. The first and second bond pad isolation structures **114** and **115** are configured to electrically isolate the bond pad **116** from the photodetectors **328** and/or the transfer transistors **320**, thereby preventing a "leakage" (i.e., a flow of current) between the bond pad **116** and the photodetectors **328** and/or the transfer transistors **320**. This increases a performance, stability, and reliability of the image sensor **300**. The first and/or second bond pad isolation structures **114**, **115** may, for example, be or comprise a dielectric material, such as silicon dioxide, silicon nitride, silicon oxynitride, or the like.

[0037] FIG. **4** illustrates a top view of some embodiments of a semiconductor structure **400** including multiple bond pads **116** laterally surrounding device regions **101***a*.

[0038] As illustrated in FIG. **4**, the semiconductor structure **400** includes multiple integrated chip die regions **402-408** spaced next to one another along a first substantially straight line **410** and a second substantially straight line **412**. When viewed in cross-section, the multiple integrated chip die regions **402-408** may be disposed along a single semiconductor substrate (e.g., a silicon substrate, such as the semiconductor substrate **110** of FIG. **1**), such that each bond pad **116** extends through the single semiconductor substrate. The first substantially straight line **410** extends along a first direction (e.g., along an x-axis) and the second substantially straight line **412** extends along a second direction (e.g., along a y-axis) perpendicular to the first direction. Each integrated chip die region **402-408** includes a device region **101***a* surrounded by a bond pad region **101***b*. In some embodiments, the device region **101***a* may include a plurality of photodetectors and/or transistors and the bond pad region **101***b* includes a plurality of bond pads **116**. The bond pads **116** may each be configured as the bond pad **116** of FIG. **1**, such that the bond pads **116** each have a reduced step height (h**1** of FIG. **1**) that is, for example, within a range of about 0 to 1.80 micrometers.

[0039] In some embodiments, during fabrication of the semiconductor structure **400**, a dicing process (e.g., performed by a mechanical saw and/or performed by a dicing laser) may be performed along the first and/or second substantially straight lines **410**, **412**. Thus, the first and second substantially straight lines **410**, **412** may be configured as scribe-lines during the dicing process. The dicing process is configured to singulate each integrated chip die region **402-408** into an individual semiconductor die, subsequently the individual semiconductor dies may be bonded to another semiconductor structure (not shown) by way of the bond pads **116**. In some embodiments, during the dicing process, conductive and/or dielectric layers and structures adjacent to each bond pad **116** may be prone to cracking and/or delamination. This may be due to a stress induced by the dicing process. However, the reduced step height (h**1** of FIG. **1**) of each bond pad **116** may mitigate and/or eliminate cracking and/or delamination of layers adjacent to the bond pads **116**. This is because the reduced step height (h**1** of FIG. **1**) of each bond pad **116** facilitates performing a bond pad opening filling process during fabrication of each bond pad **116**, such that a dielectric layer (e.g., **120** of FIG. **1**) may uniformly fill spaces and/or cracks around the bond pads **116**, thereby increasing a performance and endurance of each semiconductor die.

[0040] FIG. **5** illustrates a cross-sectional view of some embodiments of an integrated chip **500** according to alternative embodiments of the integrated chip **100** of FIG. **1**. A second dielectric layer **502** is disposed around sidewalls of the bond pad **116**. The second dielectric layer **502** is between the first dielectric layer **120** and the first passivation layer **118***a*. In some embodiments, the second dielectric layer **502** may comprise a same material as the second bond pad isolation structures **115**.

[0041] FIGS. **6-16** illustrate cross-sectional views **600-1600** of some embodiments of a method for forming a bond pad with a reduced step height and a bond pad isolation structure surrounding the bond pad according to the present disclosure. Although the cross-sectional views **600-1600** shown in FIGS. **6-16** are described with reference to a method, it will be appreciated that the structures shown in FIGS. **6-16** are not limited to the method but rather may stand alone separate of the method. Furthermore, although FIGS. **6-16** are described as a series of acts, it will be appreciated that these acts are not limiting in that the order of the acts can be altered in other embodiments, and the methods disclosed are also applicable to other structures. In other embodiments, some acts that are illustrated and/or described may be omitted in whole or in part.

[0042] As shown in cross-sectional view **600** of FIG. **6**, a semiconductor substrate **110** is provided and a shallow trench isolation (STI) structure **112** is formed on a front-side **102***f* of the semiconductor substrate **110**. In some embodiments, the semiconductor substrate **110** may, for example, be a bulk substrate (e.g., a bulk silicon substrate), a silicon-on-insulator (SOI) substrate, or some other suitable substrate. In some embodiments, a process for forming the STI structure **112** may include: selectively etching the semiconductor substrate **110** according to a masking layer (not shown) to form a trench that extends into the front-side **102***f;* and filling (e.g., by chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), thermal oxidation, etc.) the trench with a dielectric material (e.g., silicon dioxide, silicon carbide, etc.). In some embodiments, a material and/or a thickness Ts of the semiconductor substrate **110** is configured to ensure high quantum efficiency (QE) for photodetectors in NIR applications. For example, the thickness Ts of the semiconductor substrate **110** may be within a range of about 3.5 to 6 micrometers.

[0043] As shown in cross-sectional view **700** of FIG. **7**, an interconnect structure **104** is formed on the front-side **110***f* of the semiconductor substrate **110**. The interconnect structure **104** includes an interconnect dielectric structure **103**, a plurality of conductive wires **108**, and a plurality of conductive vias **106**. In some embodiments, the interconnect dielectric structure **103** may be or comprise one or more inter-level dielectric (ILD) layers. The one or more ILD layers may, for example, be or comprise an oxide, such as silicon dioxide, or another suitable oxide. In some embodiments, a process for forming the interconnect structure **104** includes forming the conductive vias **106** and the conductive wires **108** by a single damascene process or a dual damascene process. For example, a first layer of the conductive vias **106** and a first layer of the conductive wires **108** may respectively be formed by a single damascene process. Further, in such embodiments, the process includes forming remaining layers of the conductive wires **108** and the conductive vias **106** by repeatedly performing a dual damascene process. In some embodiments, the conductive wires **108** and/or the conductive vias **106** may respectively, for example, be or comprise aluminum, copper, aluminum copper, tungsten, or the like.
[0044] As shown in cross-sectional view **800** of FIG. **8**, the structure of FIG. **7** is rotated **180** degrees and the interconnect structure **104** is bonded to a carrier substrate **102**. In some embodiments, the bonding process may comprise a fusion bonding process. In some embodiments, the carrier substrate **102** may, for example, be a bulk substrate (e.g., a bulk silicon substrate), a silicon-on-insulator (SOI) substrate, or some other suitable substrate. Further, passivation layers **118***b-c* are formed on the back-side **110***b* of the semiconductor substrate **110**. In some embodiments, the passivation layers **118***b-c* may each be deposited or grown by, for example, CVD, PVD, ALD, or another suitable growth or deposition process.
[0045] As shown in cross-sectional view **900** of FIG. **9**, the structure of FIG. **8** is patterned to form an opening in the passivation layers **118***b-c* and the semiconductor substrate **110** and define an upper surface **110***us* of the semiconductor substrate **110**. The upper surface **110***us* is separated from the back-side **110***b* of the semiconductor substrate **110** by a distance ds. In some embodiments, the distance ds is within a range of about 1.2 to 3 micrometers. In some embodiments, patterning the structure of FIG. **8** may include: forming a masking layer (not shown) over the passivation layers **118***b-c;* exposing unmasked regions of the passivation layers **118***b-c* and the semiconductor substrate **110** to one or more etchants; and performing a removal process to remove the masking layer. After performing the patterning process, a first passivation layer **118***a* is formed over the semiconductor substrate **110**, thereby defining a passivation structure **118**. The first passivation layer **118***a* may, for example, be deposited or grown by CVD, PVD, ALD, or another suitable deposition or growth process. The first passivation layer **118***a* lines sidewalls and the upper surface **110***us* of the semiconductor substrate **110**. The passivation structure **118** overlies the semiconductor substrate **110** and includes the first passivation layer **118***a*, a second passivation layer **118***b*, and a third passivation layer **118***c*. The first passivation layer **118***a* may, for example, be or comprise an oxide, such as silicon dioxide, a low-k dielectric material, or the like. The first passivation layer **118***a* may directly contact the bond pad **116**. The second passivation layer **118***b* may, for example, be or comprise an oxide, such as silicon dioxide, a low-k dielectric material, or the like. The third passivation layer **118***c* may, for example, be or comprise a high-k dielectric material, or some other suitable dielectric material.
[0046] As shown in cross-sectional view **1000** of FIG. **10**, the structure of FIG. **9** is patterned, thereby defining a bond pad isolation opening **1002** and bond pad protrusion openings **1004**. In some embodiments, the patterning process includes: forming a masking layer (not shown) over the first passivation layer **118***a;* exposing unmasked regions of the first passivation layer **118***a* and the semiconductor substrate **110** to one or more etchants, thereby defining the bond pad isolation opening **1002** and the bond pad protrusion openings **1004**; and performing a removal process to remove the masking layer. In some embodiments, the patterning process may over-etch and remove at least a portion of the STI structure **112**. In other embodiments, the patterning process may stop at

a top surface of the STI structure **112**, wherein the patterning process docs not remove a portion of the STI structure **112** (not shown).

[0047] As shown in cross-sectional view **1100** of FIG. **11**, a bond pad isolation layer **1102** is formed over the passivation structure **118** and the semiconductor substrate **110**. In some embodiments, the bond pad isolation layer **1102** may, for example, be deposited or grown by PVD, CVD, ALD, thermal oxidation, or some other suitable growth or deposition process. The bond pad isolation layer **1102** may, for example, be or comprise a dielectric material, such as silicon dioxide, silicon nitride, silicon oxynitride, or the like. In some embodiments, the bond pad isolation layer **1102** may completely fill the bond pad isolation opening (**1002** of FIG. **10**) and may line each of the bond pad protrusion openings **1004**.

[0048] As shown in cross-sectional view **1200** of FIG. **12**, a patterning process is performed on the bond pad isolation layer (**1102** of FIG. **11**), thereby defining a first bond pad isolation structure **114**, second bond pad isolation structures **115**, a second dielectric layer **502**, and a bond pad opening **1202**. The patterning process further removes a portion of the STI structure **112** and a portion of the interconnect dielectric structure **103**, and exposes an upper surface of an upper conductive wire layer **108***a*. In some embodiments, the patterning process includes performing a blanket dry etch and/or a wet etch.

[0049] As shown in cross-sectional view **1300** of FIG. **13**, a bond pad layer **1302** is formed over the structure of FIG. **12** and fills at least a portion of the bond pad opening (**1202** of FIG. **12**). In some embodiments, the bond pad layer **1302** may, for example, be deposited and/or grown by electroless plating, electroplating, sputtering, or some other suitable deposition process. In some embodiments, the bond pad layer **1302** may, for example, be or comprise aluminum, copper, aluminum copper, or the like. The bond pad layer **1302** may comprise a same material, for example, as the conductive vias **106** and/or the conductive wires **108**.

[0050] As shown in cross-sectional view **1400** of FIG. **14**, the bond pad layer (**1302** of FIG. **13**) is patterned, thereby defining a bond pad **116** and a conductive ring structure **113**. A top surface of the bond pad **116** and/or the conductive ring structure **113** may each be vertically separated from the back-side **110***b* of the semiconductor substrate **110** by a step height h**1**. The bond pad **116** and/or the conductive ring structure **113** may each, for example, be or comprise aluminum, copper, aluminum copper, or the like. In some embodiments, the bond pad **116** is laterally offset from the conductive ring structure **113** by a non-zero distance. In further embodiments, patterning the bond pad layer (**1302** of FIG. **13**) includes: forming a masking layer (not shown) over the bond pad layer (**1302** of FIG. **13**); exposing unmasked regions of the bond pad layer (**1302** of FIG. **13**) to one or more etchants, thereby defining the bond pad **116** and the conductive ring structure **113**; and performing a removal process to remove the masking layer.

[0051] As shown in cross-sectional view **1500** of FIG. **15**, a first dielectric layer **120** is formed over the bond pad **116**. The first dielectric layer **120** may, for example, be or comprise silicon dioxide, some other suitable dielectric, or the like. In some embodiments, the first dielectric layer **120** may, for example, be deposited or grown by CVD, PVD, ALD, or some other suitable growth or deposition process. In some embodiments, the first dielectric layer **120** is formed with an initial thickness Ti. Subsequently, a planarization process (e.g., a chemical mechanical planarization (CMP) process) may be performed into the first dielectric layer **120** until an upper surface of the first passivation layer **118***a* is reached. In such embodiments, after the planarization process the first dielectric layer **120** may have a thickness Td that is less than the initial thickness Ti. In further embodiments, the first dielectric layer **120** may have a substantially flat upper surface that is vertically aligned with an upper surface of the first passivation layer **118***a*. In some embodiments, after performing the planarization process, a BCFA process may be performed, thereby forming light filters over the semiconductor substrate **110** in a region laterally offset from the bond pad **116** (e.g., see FIG. **3**).

[0052] In some embodiments, the thickness Td is approximately 5, 10, or 15 or more times greater than the step height h**1**. Because the step height h**1** is substantially smaller than the thickness Td of the first dielectric layer **120**, the first dielectric layer **120** may be formed in such a manner that gaps around the bond pad **116** are uniformly filled. Further, the planarization process ensures the upper surface of the first dielectric layer **120** above the bond pad **116** is substantially flat (e.g., within the tolerance of a CMP process). This, in turn, further prevents cracking and/or delamination of the first dielectric layer **120** and/or adjacent layers or structures (e.g., the second dielectric layer **502**, the bond pad **116**, the conductive ring structure **113**, etc.) during a subsequent dicing process (not shown). In such embodiments, delamination and/or cracking may be further mitigated if the bond pad **116** is in close proximity to a scribe-line (e.g., see FIG. **4**).

[0053] In some embodiments, the bond pad **116** may, for example, have been formed within a bond pad region (e.g., **101***b* of FIG. **3**) that is laterally adjacent to a device region (e.g., **101***a* of FIG. **3**) (not shown). In some embodiments, a plurality of photodetectors (e.g., **328** of FIG. **3**) may be formed within the semiconductor substrate **110** by a selective ion implant process within the device region before forming the interconnect structure **104** (not shown). Further, one or more semiconductor devices (e.g., **320** of FIG. **3**) may be formed on the front-side **110***f* of the semiconductor substrate **110**, within the device region, by one or more etching processes and one or more deposition processes before forming the interconnect structure **104** (not shown). In further embodiments, after forming the first dielectric layer **120**, a grid structure (e.g., **329** of FIG. **3**) may be formed in the device region, such that the grid structure overlies the photodetectors (not shown). A process for forming the grid structure may, for example, include: performing one or more deposition processes (e.g., CVD, PVD, ALD, etc.) to deposit one or more grid layers over the plurality of photodetectors (not shown); and performing a patterning process according to a masking layer, where the patterning process defines a plurality of color filter openings (not shown) within the device region and directly overlying a corresponding photodetector (not shown). In yet further embodiments, the patterning process may over-etch and remove at least a portion of the first passivation layer **118***a* within the device region (not shown).

[0054] As shown in cross-sectional view **1600** of FIG. **16**, a dielectric protection layer **334** is formed over the upper surface of the first dielectric layer **120**. In some embodiments, the dielectric protection layer **334** may be deposited or grown by, for example, CVD, PVD, ALD, or some other suitable growth or deposition process. After forming the dielectric protection layer **334**, a patterning process may be performed on the first dielectric layer **120** and the dielectric protection layer **334**, thereby exposing an upper surface of the bond pad **116** between bond pad openings **1160**. After performing the patterning process, an electrical connector pad **122** is formed over the upper surface of the bond pad **116**. Subsequently, the bond pad **116** may be electrically coupled to an external integrated chip (not shown) by way of a conductive wire **124**.

[0055] In some embodiments, the dielectric protection layer **334** at least partially lines the plurality of color filter openings within the device region (not shown). Further, the dielectric protection layer **334** may directly contact the first passivation layer **118***a* within the device region (e.g., see FIG. **3**). In further embodiments, a plurality of light filters (e.g., **336** of FIG. **3**) may be formed within the color filter openings (not shown). In such embodiments, the light filters may be deposited by CVD, PVD, or another suitable deposition process. In yet further embodiments, a plurality of micro-lenses (e.g., **338** of FIG. **3**) are formed over the plurality of light filters (not shown). In some embodiments, the micro-lenses may be formed by depositing a micro-lens material on the light filters (e.g., by a spin-on method or a deposition process) (not shown). A micro-lens template (not shown) having a curved upper surface is patterned above the micro-lens material. The micro-lenses are then formed by selectively etching the micro-lens material according to the micro-lens template (not shown).

[0056] FIG. **17** illustrates a method **1700** of forming a bond pad with a reduced step height and a bond pad isolation structure surrounding the bond pad according to the present disclosure. Although the method **1700** illustrates and/or described as a series of acts or events, it will be appreciated that the method is not limited to the illustrated ordering or acts. Thus, in some embodiments, the acts may be carried out in different orders than illustrated, and/or may be carried out concurrently. Further, in some embodiments, the illustrated acts or events may be subdivided into multiple acts or events, which may be carried out at separate times or concurrently with other acts or sub-acts. In some embodiments, some illustrated acts or events may be omitted, and other un-illustrated acts or events may be included. [0057] At act **1702**, a shallow trench isolation (STI) structure is formed on a front-side of a semiconductor substrate. FIG. **6** illustrates a cross-sectional view **600** corresponding to some embodiments of act **1702**. [0058] At act **1704**, an interconnect structure is formed on the front-side of the semiconductor substrate. FIG. **7** illustrates a cross-sectional view **700** corresponding to some embodiments of act **1704**. [0059] At act **1706**, a back-side of the semiconductor substrate is patterned, thereby defining an upper surface of the semiconductor substrate. FIG. **9** illustrates a cross-sectional view **900** corresponding to some embodiments of act **1706**. [0060] At act **1708**, a first passivation layer is formed over the semiconductor substrate. FIG. **9** illustrates a cross-sectional view **900** corresponding to some embodiments of act **1708**. [0061] At act **1710**, the semiconductor substrate and the first passivation layer are patterned, thereby defining a bond pad isolation opening and bond pad protrusion openings. FIG. **10** illustrates a cross-sectional view **1000** corresponding to some embodiments of act **1710**. [0062] At act **1712**, a bond pad isolation layer is formed over the semiconductor substrate. The bond pad isolation layer fills the bond pad isolation opening and lines the bond pad protrusion openings. FIG. **11** illustrates a cross-sectional view **1100** corresponding to some embodiments of act **1712**. [0063] At act **1714**, the bond pad isolation layer, the STI structure, and the interconnect structure are etched, thereby defining a bond pad opening and defining first and second bond pad isolation structures. FIG. **12** illustrates a cross-sectional view **1200** corresponding to some embodiments of act **1714**. [0064] At act **1716**, a bond pad is formed in at least a portion of the bond pad opening. The bond pad is separated from the semiconductor substrate by the first passivation layer and the second bond pad isolation structure. FIGS. **13** and **14** illustrate cross-sectional views **1300** and **1400** corresponding to some embodiments of act **1716**. [0065] At act **1718**, a first dielectric layer is formed over the bond pad. The first dielectric layer fills a remaining portion of the bond pad opening. FIG. **15** illustrates a cross-sectional view **1500** corresponding to some embodiments of act **1718**.

[0066] Accordingly, in some embodiments, the present disclosure relates to a bond pad extending through a semiconductor substrate and laterally offset from a device region. An upper surface of the bond pad is vertically below a back-side surface of the semiconductor substrate by a reduced step height. First and second bond pad isolation structures electrically isolate the bond pad from semiconductor devices (e.g., photodetectors, transistors, etc.) disposed within the device region.

[0067] In some embodiments, the present application provides a semiconductor structure, including a semiconductor substrate having a back-side surface and a front-side surface opposite the back-side surface, wherein an upper surface of the semiconductor substrate is vertically below the back-side surface; a bond pad extending through the semiconductor substrate, wherein the bond pad includes a conductive body over the upper surface of the semiconductor substrate and conductive protrusions extending from above the upper surface to below the front-side surface of the semiconductor substrate, and wherein a vertical distance between a top surface of the bond pad and the back-side surface of the semiconductor substrate is less than a height of the conductive protrusions; and a first bond pad isolation structure extending through the semiconductor substrate and laterally surrounding the conductive protrusions.

[0068] In some embodiments, the present application provides a semiconductor structure, including a first substrate overlying a second substrate and including a top surface vertically above an upper surface, wherein photodetectors are disposed in the first substrate; an interconnect structure disposed between the first substrate and the second substrate, wherein an upper conductive wire layer is disposed within the interconnect structure; a bond pad overlying the upper surface of the first substrate and extending through the first substrate to the interconnect structure, wherein the bond pad contacts the upper conductive wire layer and is laterally offset from the photodetectors, and wherein a vertical distance between the top surface of the first substrate and a top surface of the bond pad is less than a height of the bond pad overlying the upper surface of the first substrate; a first bond pad isolation structure disposed within the first substrate and laterally surrounding outer sidewalls of the bond pad; and second bond pad isolation structures disposed within the first substrate, wherein the second bond pad isolation structures directly contact sidewalls of the bond pad that extend through the first substrate.

[0069] In some embodiments, the present application provides a method for forming a semiconductor structure, the method includes forming a shallow trench isolation (STI) structure on a front-side surface of a substrate; forming an interconnect structure on the front-side surface of the substrate, wherein the interconnect structure includes a conductive wire layer; patterning the substrate to define an upper surface of the substrate vertically below a back-side surface of the substrate, wherein the back-side surface is opposite the front-side surface; forming a first passivation layer over the back-side surface of the substrate; patterning the first passivation layer and the substrate to define a bond pad isolation opening and bond pad protrusion openings, wherein the patterning of the first passivation layer and the substrate exposes an upper surface of the STI structure; depositing a bond pad isolation layer over the substrate, wherein the bond pad isolation layer fills the bond pad isolation opening and lines the bond pad protrusion openings; etching the bond pad isolation layer and the STI structure to expose an upper surface of the conductive wire layer and define first and second bond pad isolation structures; and forming a bond pad over the conductive wire layer.

[0070] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. An integrated chip, comprising: a substrate comprising an upper surface vertically below a top surface; a conductive structure in the substrate, wherein the conductive structure comprises a first segment overlying the upper surface of the substrate and one or more vertical segments under the first segment; a first isolation structure in the substrate and on opposing sides of the conductive structure; and a second isolation structure in the substrate and around the one or vertical segments, wherein a top surface of the first isolation structure and a top surface of the second isolation structure are vertically below the top surface of the substrate, wherein a bottom surface of the first isolation structure and a bottom surface of the second isolation structure are vertically above a bottom surface of the substrate.

2. The integrated chip of claim 1, wherein a first vertical distance between the top surface of the substrate and the top surface of the first isolation structure is greater than a second vertical distance between the bottom surface of the first isolation structure and the bottom surface of the substrate.

3. The integrated chip of claim 1, wherein the top surface of the second isolation structure contacts a bottom surface of the first segment, wherein outer sidewalls of the second isolation structure are spaced between outer sidewalls of the first segment.

4. The integrated chip of claim 1, wherein the top surface of the first isolation structure and the top surface of the second isolation structure are vertically above the upper surface of the substrate.

5. The integrated chip of claim 1, further comprising: a third isolation structure in the substrate and underlying the first segment, wherein the third isolation structure has a top surface above the bottom surface of the substrate, wherein the first isolation structure comprises a first sidewall facing a first direction away from the conductive structure, wherein the third isolation structure comprises a first sidewall laterally offset from the first sidewall of the first isolation structure in the first direction.

6. The integrated chip of claim 1, wherein the one or more vertical segments comprises a first vertical segment laterally offset from a second vertical segment, wherein the second isolation structure comprises a pair of inner opposing sidewalls under the first segment and laterally between the first vertical segment and the second vertical segment, wherein a first lateral distance between the pair of inner opposing sidewalls is less than a width of the first vertical segment.

7. The integrated chip of claim 6, wherein a second lateral distance between an outer sidewall of the second isolation structure and an inner sidewall of the first isolation structure is less than the first lateral distance.

8. The integrated chip of claim 1, wherein the substrate comprising a pair of opposing sidewalls extending from the top surface of the substrate to the upper surface of the substrate, wherein the first isolation structure is spaced between the pair of opposing sidewalls of the substrate.

9. An integrated chip, comprising: a conductive structure in a substrate, wherein the conductive structure comprises a body segment, a first vertical segment, and a second vertical segment, wherein the body segment overlies an upper surface of the substrate, and wherein the first and second vertical segments underlie the body segment; an outer isolation structure in the substrate and laterally wrapped around the first and second vertical segments; and a first inner isolation structure and a second inner isolation structure in the substrate and spaced between inner sidewalls of the outer isolation structure, wherein the first inner isolation structure continuously extends along and contacts outer sidewalls of the first vertical segment, and wherein the second inner isolation structure continuously extends along and contacts outer sidewalls of the second vertical segment.

10. The integrated chip of claim 9, wherein a top surface of the outer isolation structure is aligned with a top surface of the first inner isolation structure and a top surface of the second inner isolation structure.

11. The integrated chip of claim 9, wherein a first outer sidewall of the outer sidewalls of the first vertical segment faces a first direction and is laterally offset from a first outer sidewall of the body segment facing the first direction by a first lateral distance, wherein a second lateral distance between the first vertical segment and the second vertical segment is greater than the first lateral distance.

12. The integrated chip of claim 9, wherein a shortest lateral distance between the first inner isolation structure and the second inner isolation structure is greater than a shortest lateral distance between the first inner isolation structure and the outer isolation structure.

13. The integrated chip of claim 9, further comprising: a shallow trench isolation (STI) structure in the substrate and under the body segment, wherein the STI structure continuously laterally extends between the first inner isolation structure and the second inner isolation structure.

14. The integrated chip of claim 9, further comprising: a recess in the substrate, wherein the recess is defined by opposing sidewalls and the upper surface of the substrate; and a passivation layer lining the recess, wherein opposing sidewalls of the passivation layer contact the opposing sidewalls of the substrate, wherein the outer isolation structure is spaced between the opposing sidewalls of the passivation layer.

15. The integrated chip of claim 9, wherein a height of the outer isolation structure, a height of the first inner isolation structure, and a height of the second inner isolation structure are each less than a height of the substrate.

16. An integrated chip, comprising: a conductive pad in a substrate, wherein the conductive pad comprises a first segment overlying a lateral surface of the substrate and one or more vertical segments under the first segment; a first isolation structure in the substrate and laterally wrapped around the one or more vertical segments; and a conductive element over the lateral surface of the substrate and arranged on opposing sides of the first segment.

17. The integrated chip of claim 16, wherein a bottom surface of the first segment and a bottom surface of the conductive element are vertically below a top surface of the substrate.

18. The integrated chip of claim 16, wherein a top surface of the first segment is aligned with a top surface of the conductive element.

19. The integrated chip of claim 16, wherein the substrate comprises a pair of opposing sidewalls extending from a top surface of the substrate to the lateral surface of the substrate, wherein the conductive element is spaced laterally between the pair of opposing sidewalls of the substrate.

20. The integrated chip of claim 16, wherein at least a portion of the first isolation structure is spaced laterally between an inner perimeter of the conductive element and an outer perimeter of the first segment.
